# Çağatay Kılınç

## AI / ML Engineer · LLM & Agentic Systems

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### SUMMARY

AI/ML Engineer who takes LLM systems from prototype to production. **10+ years of Python** and **six years working with data**, starting in customer analytics and predictive modelling, then **four years building RAG and agentic systems**. Deep specialization in **LangGraph**: stateful graph architectures, conditional edges and human-in-the-loop workflows that keep a human approver on every high-impact action. Creator of **pdfXtractor** (16★ on GitHub) and **LeakLens**, both open source. Owns the whole path, from retrieval design (**FAISS, Milvus**) and **OpenAI / Anthropic (Claude)** integrations through FastAPI services to GCP and Azure deployment. The data-science years are what make the AI work hold up: retrieval quality, model behaviour and edge cases get **measured, not guessed**.

### CORE SKILLS

**LLM & Agentic Systems:** LangGraph (StateGraph, conditional edges, human-in-the-loop), LangChain, LlamaIndex, CrewAI, ReAct, MCP, autonomous & multi-agent systems, prompt engineering, fine-tuning (LoRA, PEFT)

**RAG & Retrieval:** RAG pipeline design, vector databases (FAISS, Milvus), hybrid & semantic search, embeddings, chunking strategies, document parsing (Camelot, agentic OCR)

**Models & APIs:** OpenAI (GPT-4o), Anthropic Claude, Llama 3, Gemini, Ollama, Hugging Face, OpenRouter

**Engineering & Deployment:** Python, FastAPI, Flask, REST API design, Streamlit, React, Google Cloud Platform, Microsoft Azure, Git

**Data Science & ML:** Pandas, NumPy, Scikit-learn, PyTorch, XGBoost, LightGBM, CatBoost, feature engineering, model evaluation, customer segmentation, churn & retention modelling, recommender systems

**Analytics & Delivery:** SQL, data modelling, experiment & result analysis, stakeholder-facing communication, technical mentoring

### PROFESSIONAL EXPERIENCE

**Freelance AI / ML Engineer** — **2022 – Present**

*Remote · clients across e-commerce, fintech and document automation*

- **LeakLens (open source):** designed and released a local-first CLI built on **LangGraph StateGraph** that detects forgotten recurring subscriptions, silent price increases and duplicate charges in bank and credit-card statements. Combines deterministic pandas-based detection with a BYOK LLM layer (OpenRouter) for merchant normalization and email drafting; currently extending it with PyPI packaging and an **MCP server** interface. *github.com/klncgty/Leaklens*
- Built a **LangGraph customer-service escalation agent** for e-commerce clients, using conditional edges plus human-in-the-loop review to route priority-tiered escalations across Slack, email and WhatsApp/SMS (Twilio), keeping a human approver on every high-impact action.
- **Project BEAM** (Italy-based client): built an end-to-end **RAG pipeline**, converting PDFs to structured JSON with Camelot, retrieving through Ragie.ai, and running local vector search with **FAISS and Milvus** for similarity search over large document sets.
- Delivered a **multi-agent chatbot** (GPT-4o, Claude 3.5 Sonnet, LangChain), a **PDF-RAG agent** (Llama 3) and a **multi-modal AI assistant** (Hugging Face, LlamaIndex), each shipped with a FastAPI backend and a Streamlit interface.

**AI Developer & Data Science Mentor, Miuul** — **2023 – 2024**

*Remote · Sweden-based EdTech company*

- Built **OpenAI/LangChain-based chatbots** and an AI-powered **course recommendation system** for the learning platform.
- Developed **Discord moderation bots** that automatically detect and prevent spam and harassment across community channels.
- Delivered data science training and 1:1 technical mentorship to learners; produced internal analysis tooling and ML models for the team.

**Data Scientist, Budget CRM** **2020 – 2022**

*Istanbul, Türkiye · customer data & CRM analytics*

- Worked end to end on **customer data**: built segmentation and behavioural analyses in Python and SQL that shaped how sales and marketing prioritised accounts.
- Developed **predictive models on CRM data** for retention and churn signals, engineering features from raw activity logs into scores the business could act on.
- Owned the analysis and reporting layer, automating recurring work with **pandas and scikit-learn** and cutting turnaround on stakeholder requests.

## OPEN SOURCE & SELECTED PROJECTS

**pdfXtractor (open source, 16★ on GitHub).** Agentic OCR engine that pulls structured tables and text out of complex PDF documents; the open-source core behind octro.io. Agents · OCR · RAG. *github.com/klncgty/pdfXtractor*

**octro.io (2025).** Productised web app built on top of pdfXtractor: extracts complex tables from PDFs into clean CSV/JSON, with an agent-based interface for querying the extracted data. *octro.io*

**RevengeTrade.** Automated cryptocurrency trading system integrating the Binance and Llama APIs for market data, signal generation (RSI, MACD, EMA, ATR) and risk management; deployed live on Microsoft Azure. *github.com/klncgty/RevengeTrade*

**Discord-Q-Analyzer.** Generative AI application that retrieves Discord messages, detects questions and analyses which topics generate the most questions. LangChain · OpenAI

## EDUCATION & CERTIFICATIONS

- **B.Sc. Physics**, Izmir Institute of Technology
- **Generative AI with Large Language Models**, DeepLearning.AI / Coursera (2024)
- **Building Applications with Vector Databases**, DeepLearning.AI (2024) · semantic search, RAG, hybrid search, recommender systems
- **Data Science & Machine Learning Program**, Miuul (2022)
- **PyTorch for Deep Learning**, Zero To Mastery Academy (2022)

## LANGUAGES

Turkish (Native) · English (Professional working proficiency) · Spanish (Beginner)